

Introduction to Anatomy Ontologies

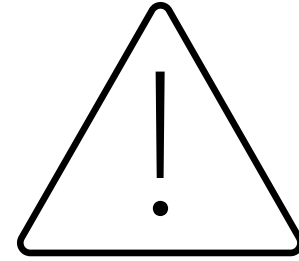
Jennifer C. Girón

Curator of Invertebrate Zoology

Museum at Texas Tech University

Outline

- General anatomy concepts
- Why ontologies
- What is an ontology
- Hierarchies
- Knowledge Graphs
- Anatomy ontologies
 - Standardizations
- Interoperability



Anatomy

- Facts based on direct observation
- What you see

Morphology

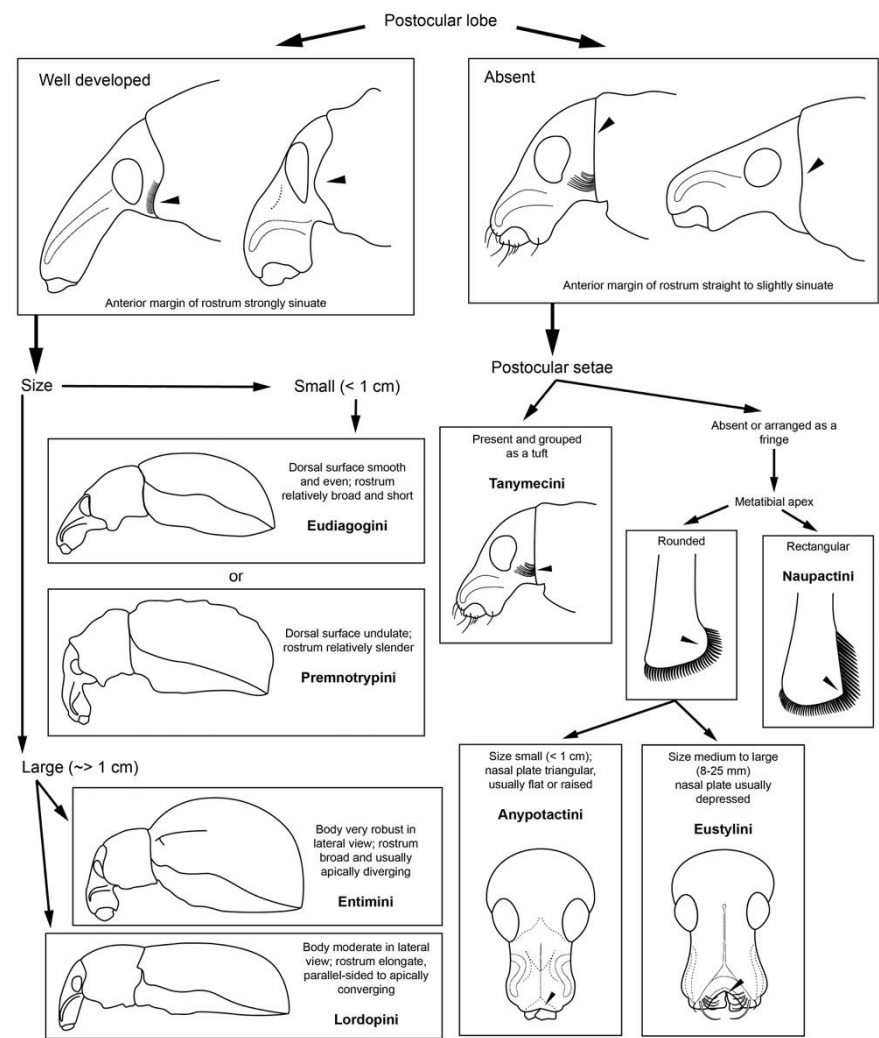
- Hypotheses to explain the facts of anatomy
 - Functional
 - Evolutionary
- What you interpret from what you see

Snodgrass, R.E. (1951) Anatomy and Morphology. *Journal of the New York Entomological Society* 59, 71–73.

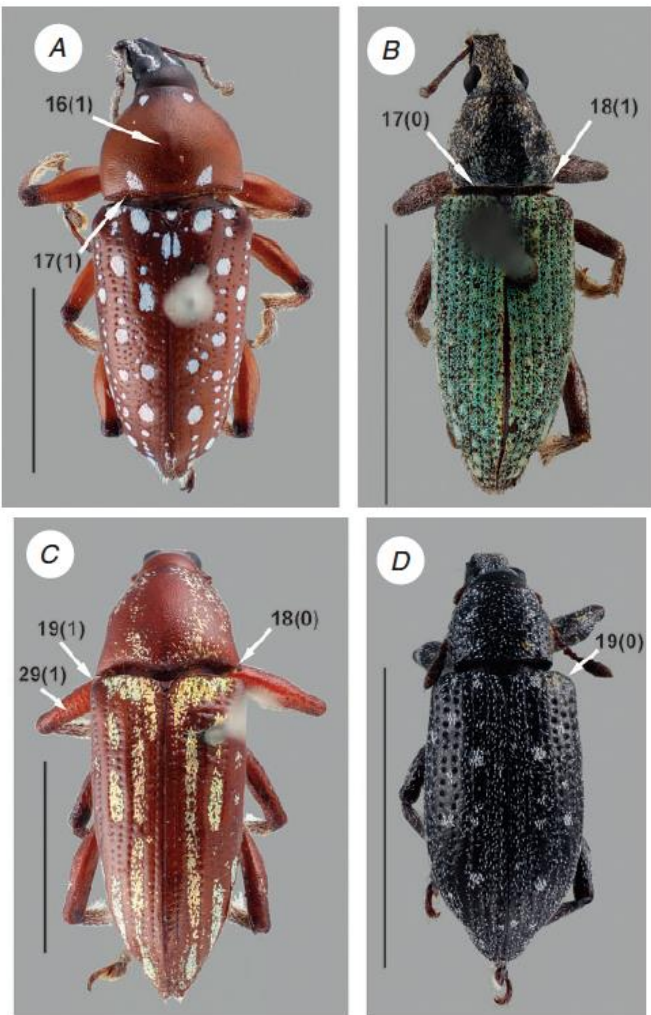
<https://www.jstor.org/stable/25005398>

Anatomical structures in Systematics

Diagnostics



Character coding



Descriptions

Genus *Katasophistes* Girón & Short, 2018
Figs 2, 5, 39, 40A–D

Katasophistes Girón & Short, 2018: 132.

Gender. Masculine.

Type species. *Katasophistes merida* Girón & Short, 2018: 136; by original designation.

Diagnosis. Medium to small beetles, body length 2.7–4.5 mm. Body shape oval to elongated in dorsal view; moderately and evenly convex in lateral view (Fig. 39). Color orange brown to dark brown, rather uniform along body regions (Fig. 39). Shape of head trapezoid. Eyes relatively small, subquadrate, at most only slightly emarginated anteriorly, moderately projected from outline of head. Clypeus trapezoid, with anterior margin broadly emarginate. Labrum fully exposed. Mentum with strong median anterior depression sometimes limited by low transverse carina; surface of mentum with lateral oblique ridges (Fig. 39C, F). Antennae with nine antennomeres; cupule slightly asymmetric, with rounded outline. Maxillary palps moderately long, $0.7 \times$ to nearly as long as width of head; inner margin of maxillary palpomere 2 slightly curved near apex, outer margin curved, sometimes strongly, along apical half (Fig. 39C, F). Each elytron with five rows of deep/large systematic punctures; elytra without sutural striae, with outer margins slightly flared; serial punctures absent (Fig. 39A, D). Prosternum slightly convex to tectiform. Posterior elevation of mesoventrite, with a well-defined, curved transverse ridge; anapleural sutures forming an obtuse angle, separated at anterior margin by distance $0.2\text{--}0.3 \times$ the width of anterior margin of mesepisternum. Metaventricle densely pubescent, except for large median rhomboid glabrous patch (Fig. 39C, F). Protibiae with spines of anterior row hair-



Natural language

- How humans communicate with each other

An ontology is a tool to
formally organize knowledge

Formal (logic) language

- Understandable by humans AND processable by machines
- Representation of terms, definitions, and relations (obo/owl)
- Simplest form – controlled vocabularies



The OBO Foundry

- Principles – Open science
- **Interoperability**
 - BFO – Basic Formal Ontology
 - UBERON – multi-species anatomy ontology
 - BSPO – Biological Spatial Ontology
 - PATO – Phenotype And Trait Ontology
 - RO – Relation Ontology

The Ontology Development Kit (ODK)

 Build the ODK images and run the tests **passing** DOI [10.5281/zenodo.4973944](https://doi.org/10.5281/zenodo.4973944)

Initialize a Git repo for managing your ontology the OBO Library way!



Allows collaborative editing

Anatomy Ontologies for Insects

Hymenoptera Anatomy Ontology Portal



go: [search](#) | [analyze](#) | [give feedback](#) | [references](#) | [terms](#) | [tree](#) | [partonomy](#) | [pulse](#) | [about / how to cite](#)

Result

URI: http://purl.obolibrary.org/obo/HAO_0000513

maxilla synonyms: [first maxilla](#), [maxilla prima](#)

The [appendage](#) that is encircled by the [area](#) that is proximally delimited by the [hypostoma](#) posteriorly, the median [margin](#) of the [mandible](#) laterally, the [labrum](#) anterolaterally and the [labium](#) medially.

written by: Miko, I. 2009. -2019 Curator. Hymenoptera Anatomy Ontology.



[- click for more detail -](#)

FlyBase Tools ▾ Downloads ▾ Links ▾ Community ▾ About ▾ Help ▾

FB2026_01, released March 12, 2026 A Database of *Drosophila* Genes & Genomes J2G ▾ Jump to Gene Go

BLAST JBrowse New to Flies? GO PHENOTYPE ANATOMY DISEASE MORE ID Validator Batch Download Fast-Track Latest Technical Advances

Citing FlyBase

External Resources

- News and Outreach
- Drosophila* Journals and Protocols
- Tool/Reagent Resources
- Meetings/Courses/Jobs
- Drosophila* in Disease Research
- Inter-species
- Fly Boards

FlyBase Funding Update – March 2026

The NHGRI funding for FlyBase has been restored for US-based FlyBase teams but the foreign subaward that supported the essential curation team at the University of Cambridge UK has been terminated. Therefore, the Cambridge UK team now requests annual contributions from fly labs to support curation of new data into FlyBase and the *Alliance of Genome Resources*. Without these annual contributions, FlyBase will become frozen in its current state and become progressively less useful.

Please visit the 'Contribute to FlyBase' wiki page for details of how to contribute.

QuickSearch

Human Disease Protein Domains Gene Groups Complexes Pathways Data Class

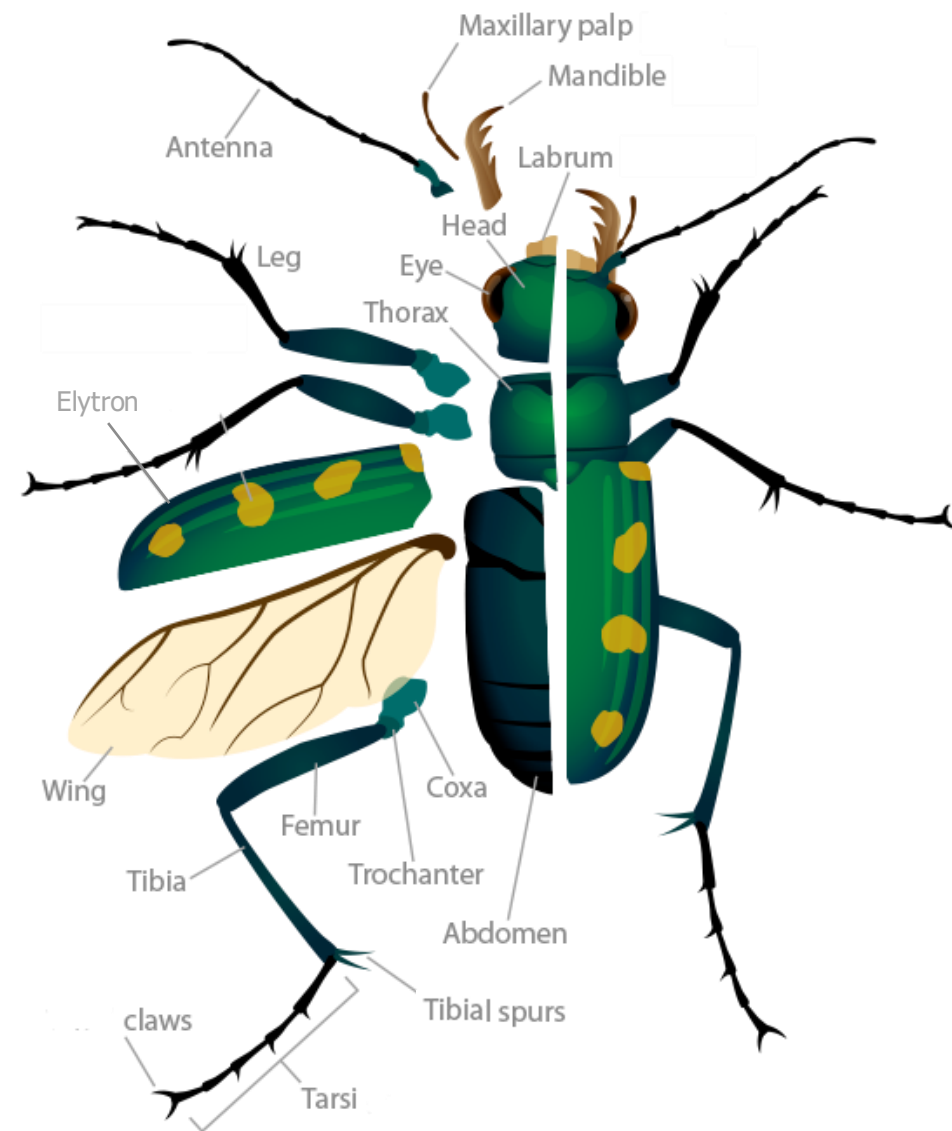
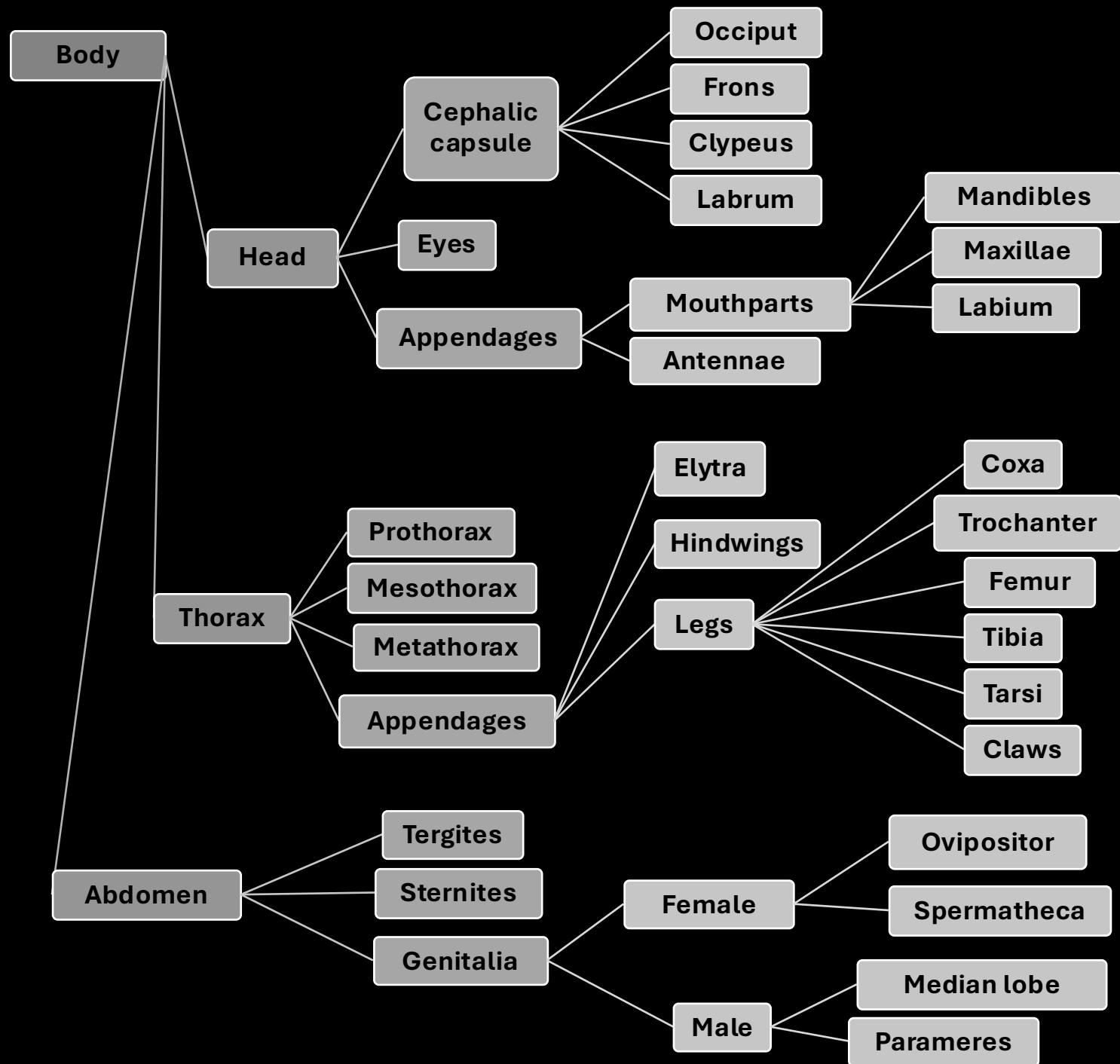
Search FlyBase GO Homologs GAL4 etc Expression Phenotype References

Everything ▾ ? Search

[Click here to submit multiple IDs/symbols.](#)

Note: Wild cards (*) can be added to your search term

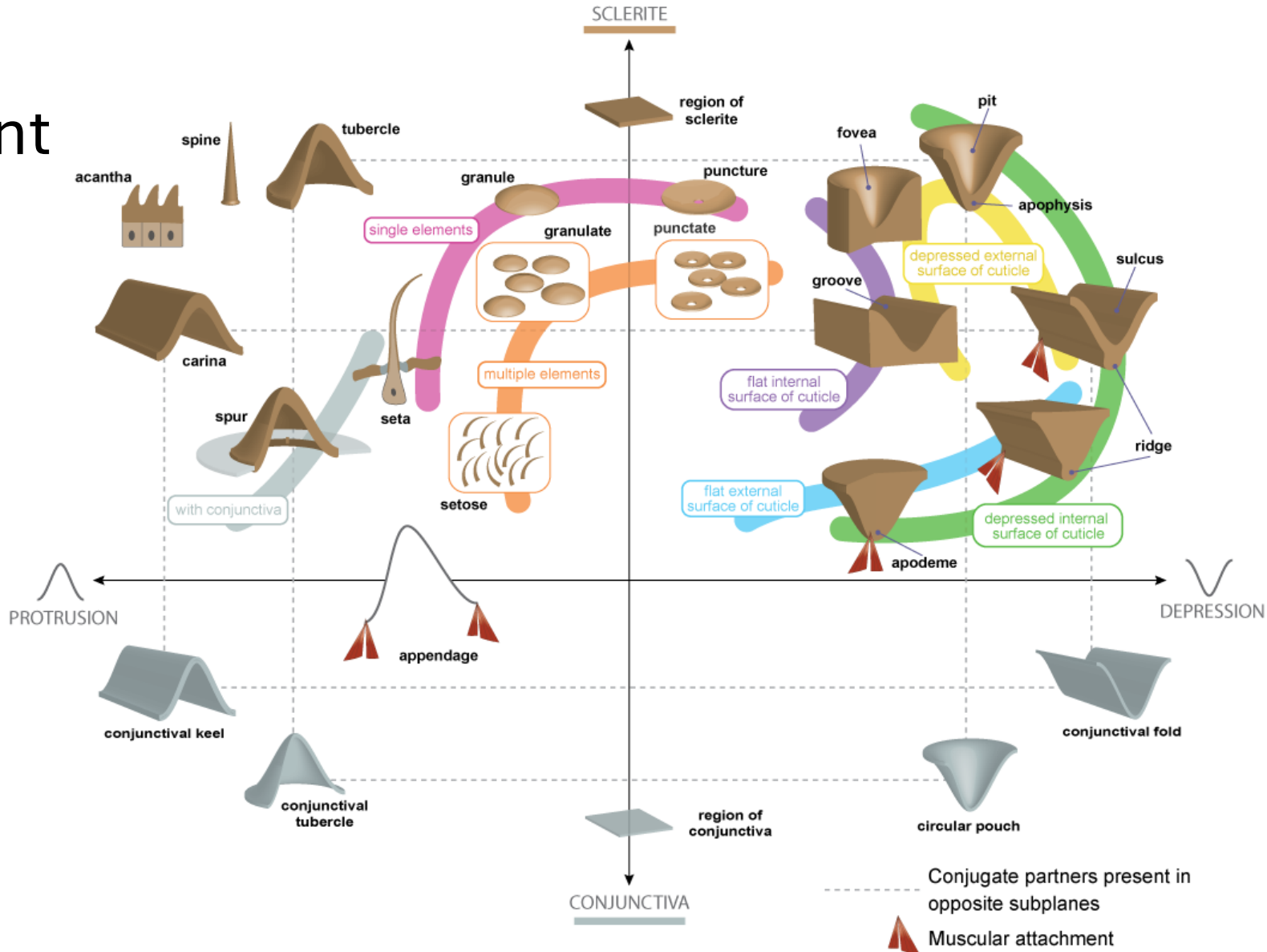
- **Ontology for the Anatomy of the Insect SkeletoMuscular system (AISM)**
- **Coleoptera Anatomy Ontology (COLAO)**
- Hymenoptera Anatomy Ontology
- *Drosophila* anatomy ontology
- *Drosophila* Phenotype Ontology
- Mosquito Gross Anatomy Ontology
- Lepidoptera Anatomy Ontology
- *Tribolium* Ontology
- Collembola Anatomy Ontology
- Spider Anatomy Ontology
- Tick Anatomy Ontology
- Ontology of Arthropod Circulatory Systems



Cuticular system - Structural properties

Kind of cuticular element

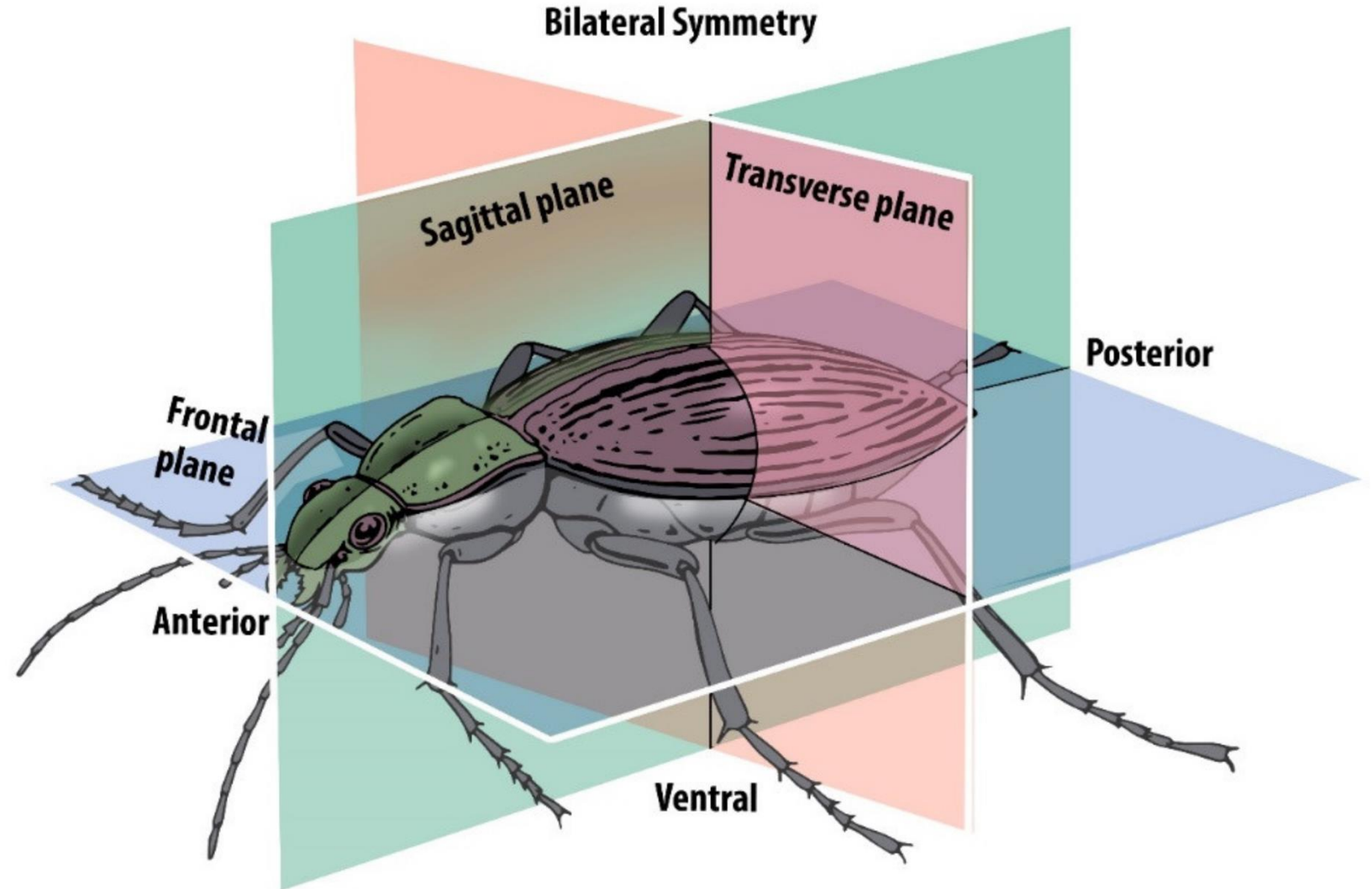
- Sclerite
 - Laminar
 - Ring-like
- Conjunctiva
 - Protrusion
 - Depression

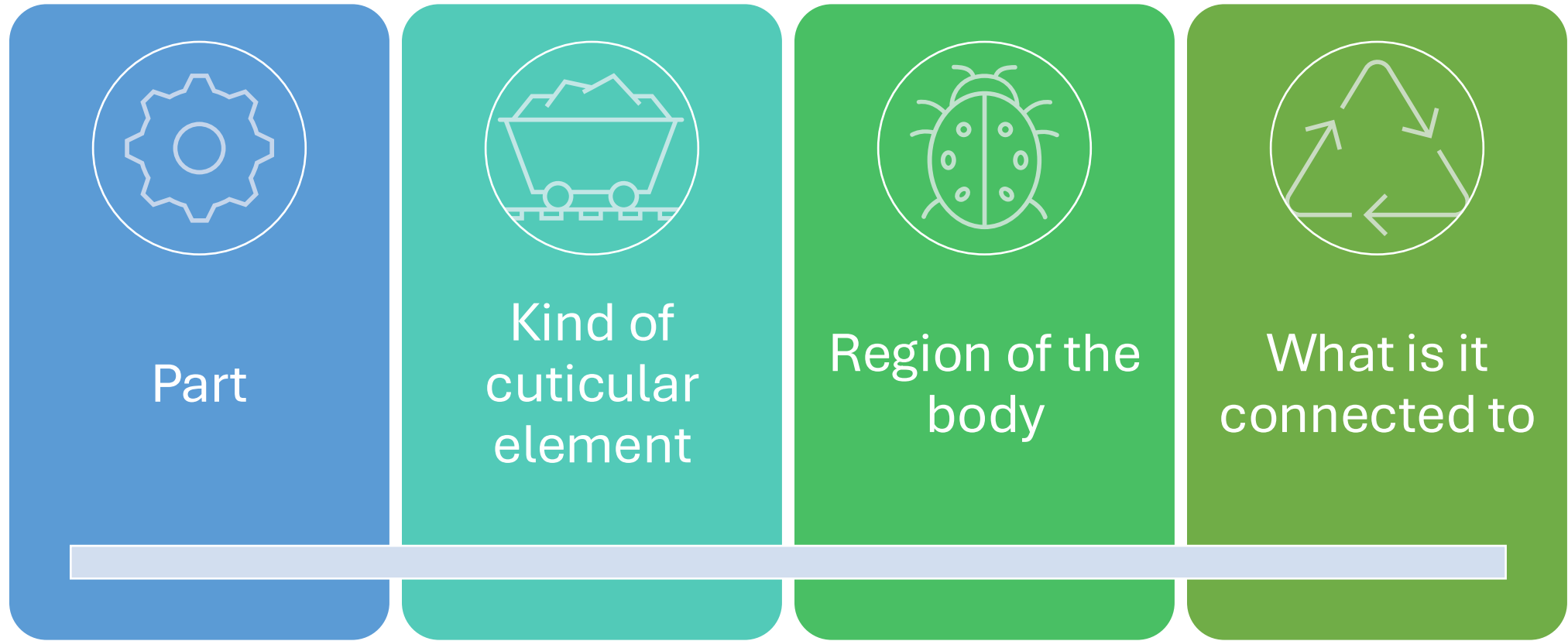


Cuticular system – Orientation – Positional relationships

Locators

- body tagma
- body region
 - Dorsal
 - Ventral
- Anterior
- Posterior
- Lateral
- Mesal





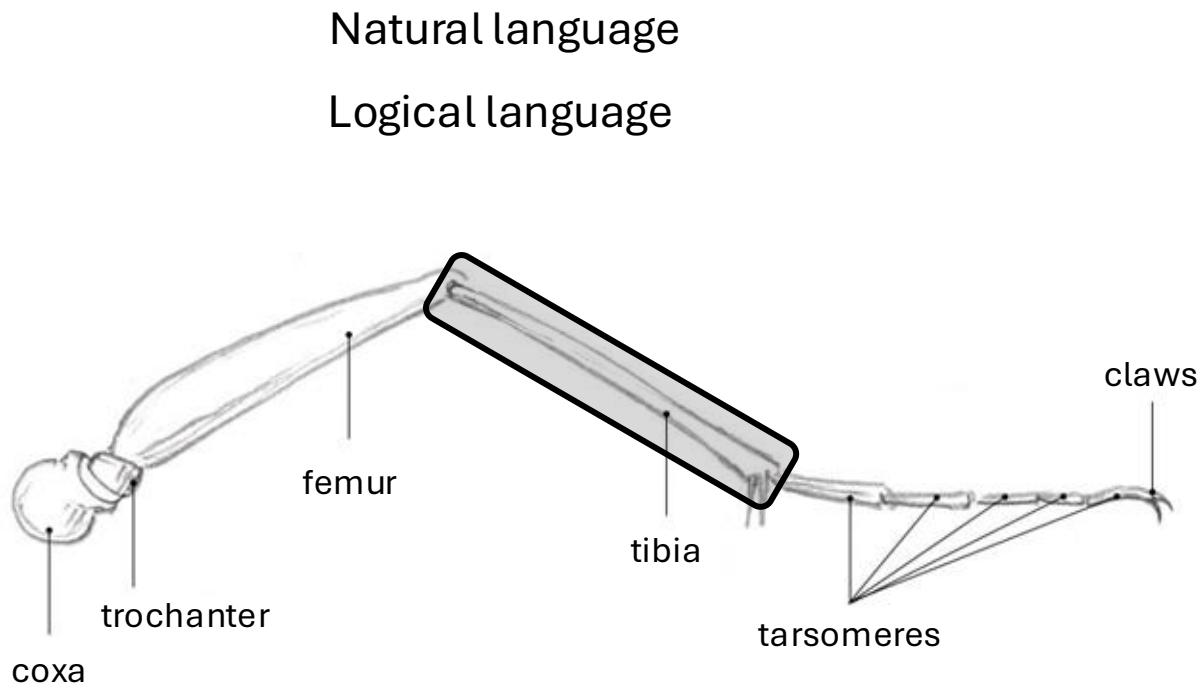
Multiple aspects

Each aspect with multiple categories

Categories can be organized hierarchically

ontology

ontology: representation of terms, definitions, and relations



<https://johnmuirlaws.com/draw-insects-understanding-drawing-legs/>

Object	Property	Subject
tibia	part of	leg
leg	has part	tibia



Part



Kind of cuticular element

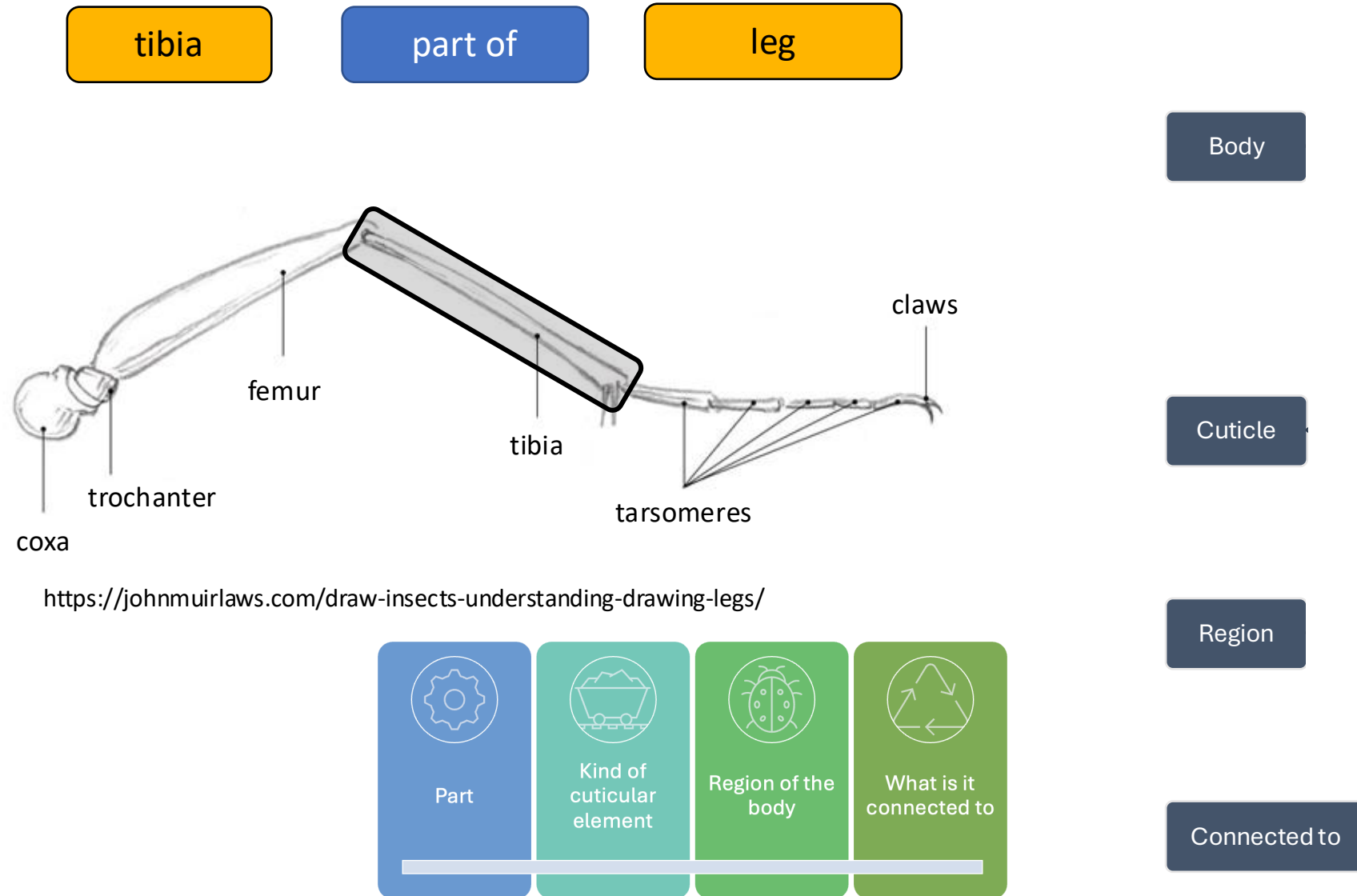


Region of the body



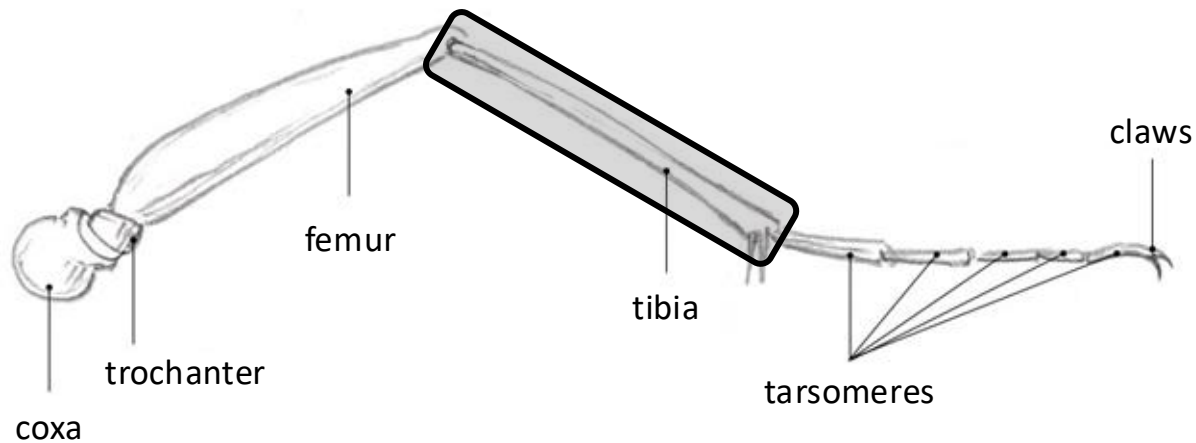
What is it connected to

Anatomy ontologies: systems of multiple hierarchies



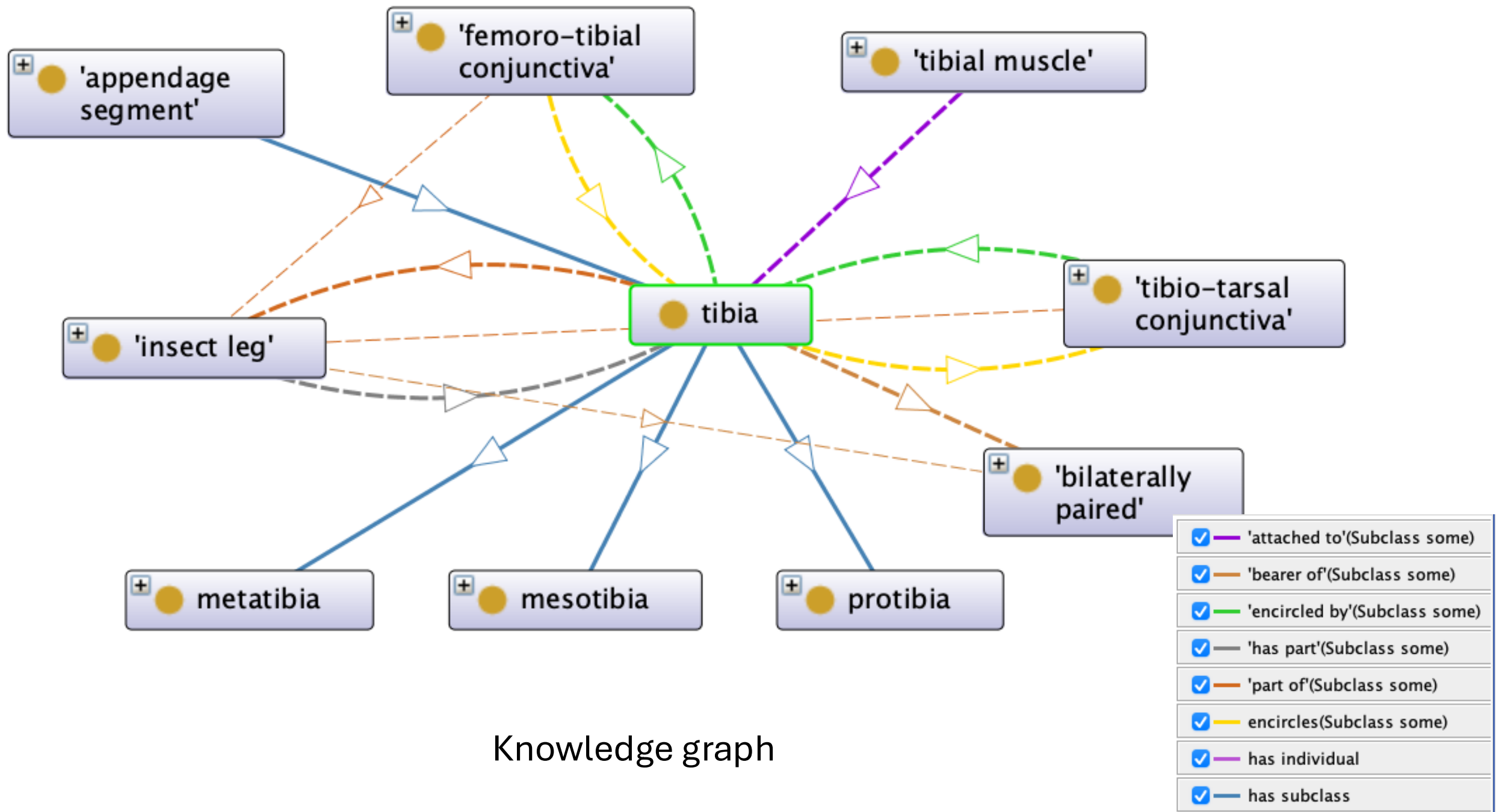
Anatomy ontologies: systems of multiple hierarchies

properties are logically associated
(inherited)



<https://johnmurlaws.com/draw-insects-understanding-drawing-legs/>

- part of the insect leg
 - part of insect thorax
- appendage segment
 - ring sclerite
 - bilaterally paired
- adjacent to the distal margin of the femur
 - encircled by the femoro-tibial conjunctiva
- adjacent to the proximal margin of the tarsus
 - encircles the tibio-tarsal conjunctiva



Tibia in Protégé

The screenshot displays the Protégé ontology editor interface. On the left, the 'Class hierarchy: tibia' panel shows a tree of classes. The 'tibia' class is highlighted in blue, and a black arrow points to it. Above it, the 'ring sclerite' class is also highlighted, with another black arrow pointing to it. The right panel shows the 'Annotations: tibia' section. A black arrow points to the URL 'http://purl.obolibrary.org/obo/AISM_0000075' in the top bar. The 'Annotations' section contains several annotations, including 'rdfs:label' (language: en) with the value 'tibia', 'sensu' (language: en) with a citation and DOI, and 'definition' (language: en) with a detailed description. Below the annotations, the 'Description: tibia' section shows a list of sub-classes, including 'adjacent to' some 'distal margin' and 'proximal margin', 'adjacent to' some 'proximal margin' and 'part of' some 'tarsus', 'appendage segment', 'encircled by' some 'femoro-tibial conjunctiva', 'has characteristic' some 'bilaterally paired', 'part of' some 'insect leg', and 'encircles' some 'tibio-tarsal conjunctiva'.

Annotation properties: Datatypes: Individuals: Classes: Object properties: Data properties: **Class hierarchy: tibia** **Annotations: tibia**

Annotations: **rdfs:label** [language: en] tibia

sensu [language: en]
Beutel RG, Friedrich F, Yang X-K, Ge S-Q (2014) Insect Morphology and Phylogeny Insect morphology and phylogeny: a textbook for students of entomology. De Gruyter.
<https://doi.org/10.1515/9783110264043>
"Tibia: 4th element of leg; usually cylindrical and equipped with paired terminal spurs, often with rows of spines."

definition [language: en]
The segment of a leg attached to the distal margin of the femur by the femoro-tibial conjunctiva and to the proximal margin of the tarsus by the tibio-tarsal conjunctiva.
contributor
jgiron <https://orcid.org/0000-0002-0851-6883>
creation_date
2021-04-03

Description: tibia

Equivalent To

SubClass Of

- 'adjacent to' some ('distal margin' and ('part of' some femur))
- 'adjacent to' some ('proximal margin' and ('part of' some tarsus))
- 'appendage segment'
- 'encircled by' some 'femoro-tibial conjunctiva'
- 'has characteristic' some 'bilaterally paired'
- 'part of' some 'insect leg'
- encircles some 'tibio-tarsal conjunctiva'

Terms and definitions must be unique within ontology

Important for reasoning

Definitions

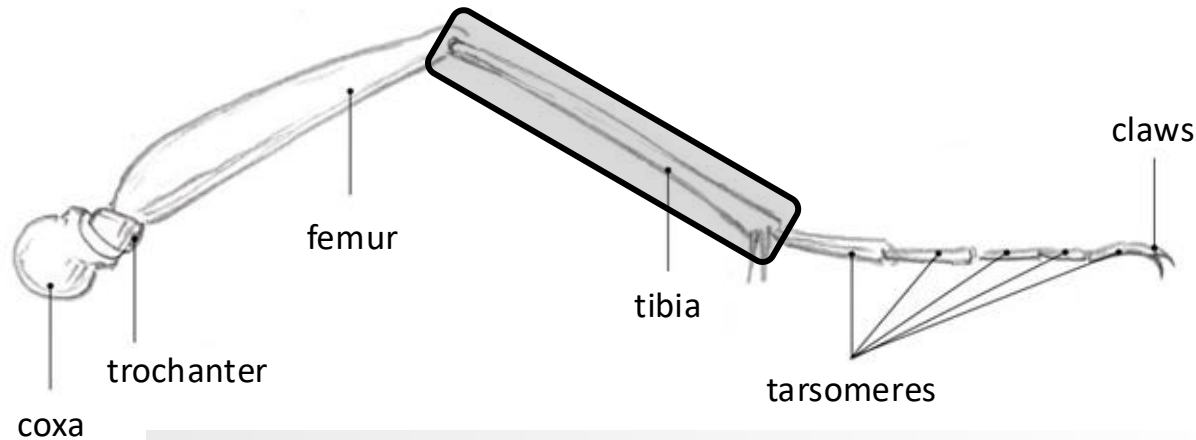
Logical

Natural

Contain the same elements

Interoperability

UBERON - multi-species anatomy ontology



tibia

http://purl.obolibrary.org/obo/AISM_0000075 Copy

Definition:

The cuticle of the segment of a leg attached to the distal margin of the femur by the femoro-tibial conjunctiva.

tibia

http://purl.obolibrary.org/obo/UBERON_0000979 Copy

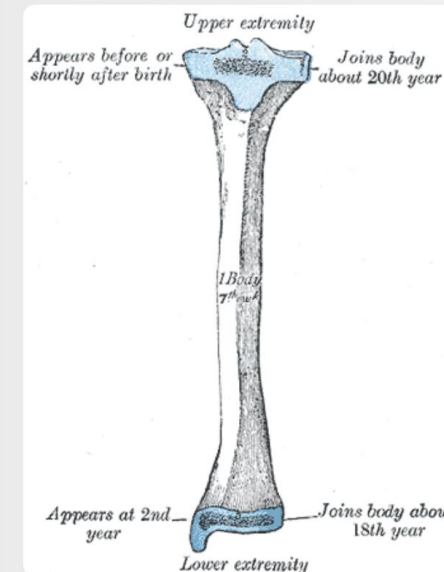
Definition:

The major preaxial endochondral bone in the posterior zeugopod[Phenoscape].

Also appears in [CL](#) [CPONT](#) [DOID](#) [EFO](#) [FOVT](#) + 10

Exact Synonyms [shankbone](#) [shinbone](#)

Depicted by



Vertebrate biased

Check that the terms and definitions you are using are appropriate

BSPO - Biological Spatial Ontology

Biological Spatial Ontology

Version 2023-05-27

An ontology for representing spatial concepts, anatomical axes, gradients, regions, planes, sides and surfaces. These concepts can be used at multiple biological scales and in a diversity of taxa, including plants, animals and fungi. The BSPO is used to provide a source of anatomical location descriptors for logically defining anatomical entity classes in anatomy ontologies.

Imports from BFO CARO DC DCTERMS GO + 8

Exports to AIMS CL CLO COLAO COVOC + 34

proximal

Jump to

proximal side

proximal to

proximal region

proximal surface

proximal margin

BSPO BSPO:0000061

BSPO BSPO:0000100

BSPO BSPO:0000077

BSPO BSPO:0000377

BSPO BSPO:0000677

Search OLS for proximal

Search

PATO - Phenotype And Trait Ontology

- quality
 - color
 - shape
 - size
 - and many other attributes
 - chemical
 - physiological
 - behavioral
- BFO - Basic Formal Ontology

Class Hierarchy

```
Thing
+ BFO\_0000020
  + quality
    + qualitative
    + process quality
    + variability
    + increased quality
    + decreased quality
    + dendrite quality
    + quantitative
    - physical object quality
      + morphology
      + physical quality
      + functionality
      + disposition
      + organismal quality
      + population quality
      + molecular quality
      + aplastic/hypoplastic
      + decreased object quality
      + increased object quality
      - structurally discontinuous
      + absence of physical object
    more...
```

RO - Relation Ontology

- has_part- BFO:0000051
- part_of- BFO:0000050
- has_characteristic- RO:0000053
- characteristic_of- RO:0000052
- increased_in_magnitude_relative_to- RO:0015007
- decreased_in_magnitude_relative_to- RO:0015008



phenoscript

Interoperability

- GO - Gene Ontology
- CL - Cell Ontology
- ChEBI - Chemical Entities of Biological Interest
- NCBI taxonomy ontology

Where to search for ontologies or terms?

 OBO Foundry About ▾ Principles ▾ Ontologies ▾ Citation ▾ Participate ▾ Newsletter ▾ FAQ

Open Biological and Biomedical Ontology Foundry

Community development of interoperable ontologies for the biological sciences

Learn about OBO best practices and community resources

- [OBO Foundry principles](#)
- [OBO tutorial](#)
- [Ontology browsers, tutorials, and tools](#)

Participate

- [Code of Conduct](#)
- Join the [OBO mailing list](#) and the [OBO Community Slack workspace](#)
- [OBO Foundry Operations and Working Groups](#)
- [Submit bug reports or suggestions for improvement via GitHub](#)
- [Submit your ontology to be considered for inclusion in the OBO Foundry](#)

Find, use, and contribute to community ontologies

Download table as: [[YAML](#) | [JSON-LD](#) | [RDF/Turtle](#)]

<https://obofoundry.org/>



Where to search for ontologies or terms?



[Home](#)[Intro](#)[Statistics](#)[SPARQL](#)[Ontobee](#)[Annotator](#)[Tutorial](#)[FAQs](#)[References](#)[Links](#)[Contact](#)[Acknowledge](#)[News](#)

Welcome to Ontobee!

Ontobee: A [linked data](#) server designed for ontologies. Ontobee is aimed to facilitate ontology data sharing, visualization, query, integration, and analysis. Ontobee dynamically [dereferences](#) and presents individual ontology term URIs to (i) *HTML web pages* for user-friendly web browsing and navigation, and to (ii) *RDF source code* for [Semantic Web](#) applications. Ontobee is the default linked data server for most [OBO Foundry library ontologies](#). Ontobee has also been used for many non-OBO ontologies.

Please select an ontology (optional) ▼

Keywords:

Search terms

Batch Search

Jump to <http://purl.obolibrary.org/obo/>

Go

<https://ontobee.org/>



<https://www.ebi.ac.uk/ols4/>



[Home](#)[Ontologies](#)[Tag Text](#)[API Docs](#)[MCP Server](#)[About](#)[Downloads](#)

obo foundry

agriculture

anatomy and development

biological systems

chemistry and biochemistry

diet, metabolomics, and nutrition

environment

health

information

information technology

investigations

microbiology

organisms

phenotype

simulation

upper

drosophilid anatomy

ms cross-linker reagents

ms experiments

major histocompatibility complex

abiotic stress

adverse events

agronomy

ancestry

artificial life

behavior

bio-repository

biobanking

biodiversity collections

biological nomenclature

biological process

biological sequence

biology

cells

clinical

clinical documentation

clinical epidemiology

cohort studies

collections of organisms

computational biology

developmental life stage

dietary surveys

digital evolution

disease

ecological functions

ecological interactions

evidence

experimental conditions

food

functional genomics

genotype-to-phenotype associations

glycan structure

imaging experiments

informed consent

insect anatomy

learning systems

life cycle stage

mathematical modeling

medical

medicine

microbiomes

numerical simulation

nutrition, nutritional studies, nutrition professionals

nutritional epidemiology

observational studies

ontology term annotation

parasite organism

pathways

plant disease

plant phenotypes

population biology

proteins

radiation biology

relations

scholarly contribution roles

scientific claims

simulation algorithms

social

software

specimens

statistics

systems biology

taxonomy

toxicity

vertebrate traits

Ontology ⌵

Filter by Ontology

×

ID ⬆

Filter by ID

×

Description ⌵

Filter by Description

×

Actions

Questions?

Jennifer C. Girón
entiminae@gmail.com

Plant ontologies

 Plant Ontology (PO)

[Search Database](#)
[External resources](#)
[Download](#)
[Request PO Documents](#)
[Release notes](#)
[Software](#)
[Publications](#)
[About](#)
[License](#)

Example Searches: type 'endosperm' or 'PO:0009089' and select 'PO terms' type: 'CONSTANS' or 'AT5G15850' and select 'Annotations'

☒ **PO terms**
☐ **Annotations**
☐ **exact match**

More options: [Browse PO](#), [Advanced Search](#) and [Help](#)

Links to: [PO Web Services](#)

- Plant Ontology (PO)
 - Plant anatomy
 - Structure and development
- Plant Traits (TO)
- Plant Phenology Ontology (PPO)

Vertebrate ontologies



Uberon Multi-species Anatomy Ontology



Uberon Ontology Documentation



Quick links:

- [Uberon on Github](#)
- [Issue Tracker](#)
- [Browse Uberon on OLS](#)
- [About Uberon](#)
- [The Uberon Team](#)

- [UBERON](#)
- [FuTRES Ontology of Vertebrate Traits \(FOVT\)](#)
- [Vertebrate Taxonomy Ontology \(VTO\)](#)